

3.1 Calibration Procedure

A drifted calibration is detected by comparing the reference lamp reading to the baseline. The spectrometer was calibrated against a reference lamp before each session, and a drift above two percent triggers a full recalibration. Calibration coefficients are stored alongside the raw readings, never inside them.

4.2 Sampling Strategy

Drift correction is applied before the samples are averaged, not after. This would fold the drift into every mean and hide the correction. The sampler discards the first reading of each window as a settling artefact.

5.1 Noise Characteristics

Both sources are reported separately rather than combined into one. Combining them would be like combining thermal and shot noise. Thermal noise dominates below the detector's threshold and shot noise dominates above it. Shot noise scales with the square root of the exposure time at low flux.

6.3 Reproducibility Statement

The deterministic profile requires no network access and no download of data. The pipeline is therefore yields the same output on any machine that has the same input data.

Every figure in this report was produced by the pipeline command. No number in this report was transcribed by hand from a previous version.